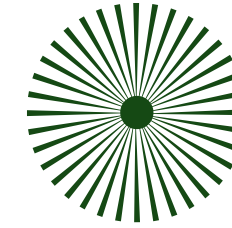


AI-DRIVEN CLIMATE-AWARE DECISION SUPPORT AND RISK FORECASTING SYSTEMS FOR SRI LANKA

PROJECT ID : R26-IT-046



INTRODUCTION



Sri Lanka is increasingly experiencing climate-related disasters, but many communities lack timely information and tools to respond effectively.

Recurring Floods

- Flooding occurs almost every year in Sri Lanka

Dengue Outbreaks

- Dengue cases are often detected only after the disease spreads widely

Challenges for Farmers

- Farmers struggle with mistimed irrigation and fertilizer application

Dangerous Road Conditions

- Roads become flooded and hazardous during heavy rain
- Increases traffic accidents, delays, and safety risks

PROBLEM



THE PROPOSED SOLUTION - SOLUTION OVERVIEW

What It Is ?

- An AI-powered climate decision support system that is built specifically for Sri Lanka's geography, climate, and communities.
- Four integrated components covering flood, disease, agriculture, and transport.

What It Does ?

- Predicts floods up to 6 hours before they occur.
- Early warning system for dengue outbreaks.
- Delivers real-time farming advisories to farmers.
- Recommends safer travel routes during adverse weather

Shortcomings of Existing Systems

Rule-Based & Reactive Systems

- Government systems like Disaster Management Centre and Department of Meteorology Sri Lanka use rule-based alerts.
- Mostly rely on historical thresholds instead of AI predictions.
- Alerts are often delayed.

Separate Systems for Each Domain

- Different platforms handle different problems.
- No single platform integrates disaster, health, agriculture, and mobility data.

Our Approach

Real-Time AI Prediction

- Uses CNN-LSTM, Transformer, and LSTM models.
- Runs on live data streams.
- Provides 0 – 6 hour flood forecasts and weekly dengue risk maps.

Integrated Smart System

- Combines multiple services in one platform:
 - Flood prediction
 - Dengue outbreak forecasting
 - Crop advisories
 - Climate-aware navigation
- Uses Sri Lanka specific datasets.

Role-Based Recommendations

- Available through mobile and web apps.
- Provides specific advice for different users
 - Farmers
 - Health workers
 - Drivers
 - Disaster management teams



IT22084422 | T.D.C.D INDUWARA

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

Problem Definition

- Flood warnings arrive too late, after heavy rain has already started.
- Existing tools are too complex for ordinary people to use.
- No single system combines rainfall, river levels and flood prediction together

Research Objectives

- Collect and preprocess historical rainfall and river level data from Sri Lankan sources.
- Analyze relationships between rainfall intensity, river levels and flood occurrences.
- Develop a predictive model to estimate future flood risks.

Novelty

- Provides hyper-local flood forecasts up to 6 hours ahead for specific locations
- Combines rainfall radar, river level data and historical flood records together in one system.
- Delivers flood maps and alerts directly to citizens through a mobile app.

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

Research Methodology

- Follows a Design Science approach, focused on building and evaluating a practical system.
- Historical rainfall, river level and flood event data collected from official Sri Lankan sources.

Latest Technologies

- CNN-LSTM - deep learning model capturing spatial & temporal flood dynamics
- TensorFlow - GPU-accelerated training & cloud deployment
- Firebase - real-time database, authentication, push notifications
- React Native - cross-platform mobile app with interactive flood maps

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

Validation & Evaluation

- Model tested against real historical flood events in Sri Lanka
- MAE and RMSE used to measure accuracy of predicted river levels
- Precision, Recall and F1-score used to evaluate flood and non-flood classifications
- Mobile app evaluated for usability, map accuracy and alert clarity

Data, Security & Ethics

- Data sourced from DMC & Department of Meteorology
- Firebase security rules & HTTPS protocols protect user & forecast data
- Responsible handling of location-sensitive information with full source acknowledgment

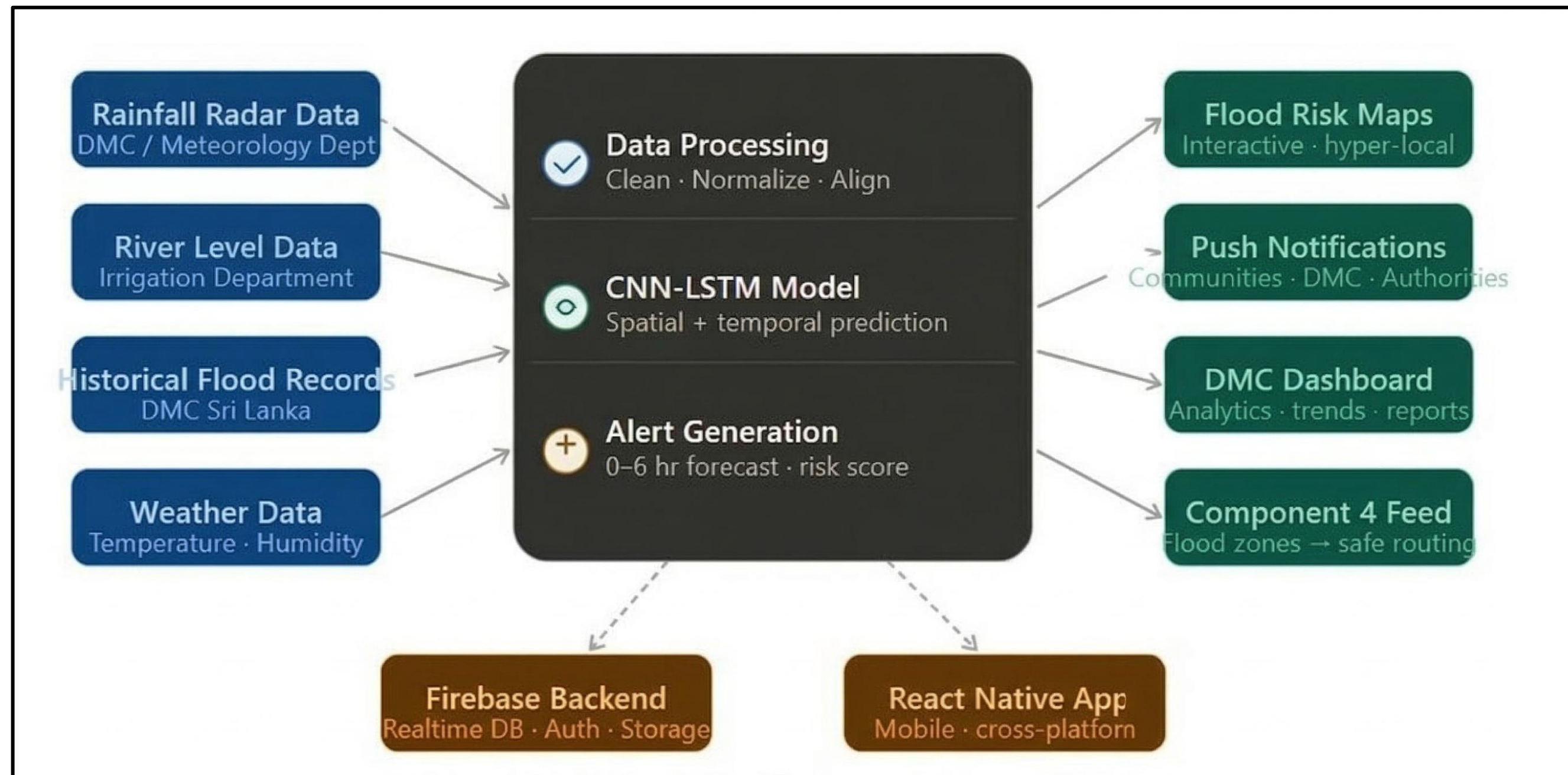
AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

Datasets Used

Dataset	Source	Used In Component	Purpose
Rainfall Data	Department of Meteorology	CNN Model	Detect spatial rainfall patterns
River Water Level Data	Disaster Management Centre	LSTM Model	Analyze river level trends
Historical Flood Event Data	Disaster Management Centre	Model Training	Identify flood occurrence patterns

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

System Overview





IT22100702 | LAKSHAN H.M.M

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

Problem definition

- Dengue is a major public health problem in Sri Lanka.
- Current systems mostly react after cases are reported.

Novelty

- Many existing studies only predict dengue case numbers.
- This research combines climate data, dengue history, risk maps, and warning alerts.
- It is designed specifically for Sri Lanka's climate and dengue pattern.

Research objectives

- Develop an AI system using climate data and dengue case data.
- Predict weekly dengue risk in different areas.
- Generate early warnings for high-risk locations.
- Help health authorities plan prevention and resource use

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

Research Methodology

- Used the Design Science Research approach to address the real-world problem of Dengue Risk Prediction.
- The model is evaluated using standard metrics to ensure accurate predictions and support early flood warnings for authorities.

Latest Technologies

- Python, Pandas, NumPy, and Scikit-learn for data processing and ML tasks.
- TensorFlow for LSTM and Transformer model development.
- Firebase and NoSQL database for storing predictions and alerts.
- React Native for mobile-based risk maps and warning notifications.

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

Validation & Evaluation

- Data is divided into training, validation, and testing sets.
- RMSE and MAE are used to measure prediction error.
- AUC is used for risk classification, and accuracy, precision, recall, and F1-score can also be reported.
- Results are compared with ARIMA and Random Forest to show improvement.
- The main goal is to check whether the system gives earlier and more useful warnings.

Data, Security & Ethics

- Use public or properly approved datasets only.
- Keep health data grouped by area so no person can be identified.
- Protect stored data with authentication, access control, and secure storage.
- Predictions should support decisions, not replace health experts.
- Results should be clear, responsible, and checked regularly for reliability.

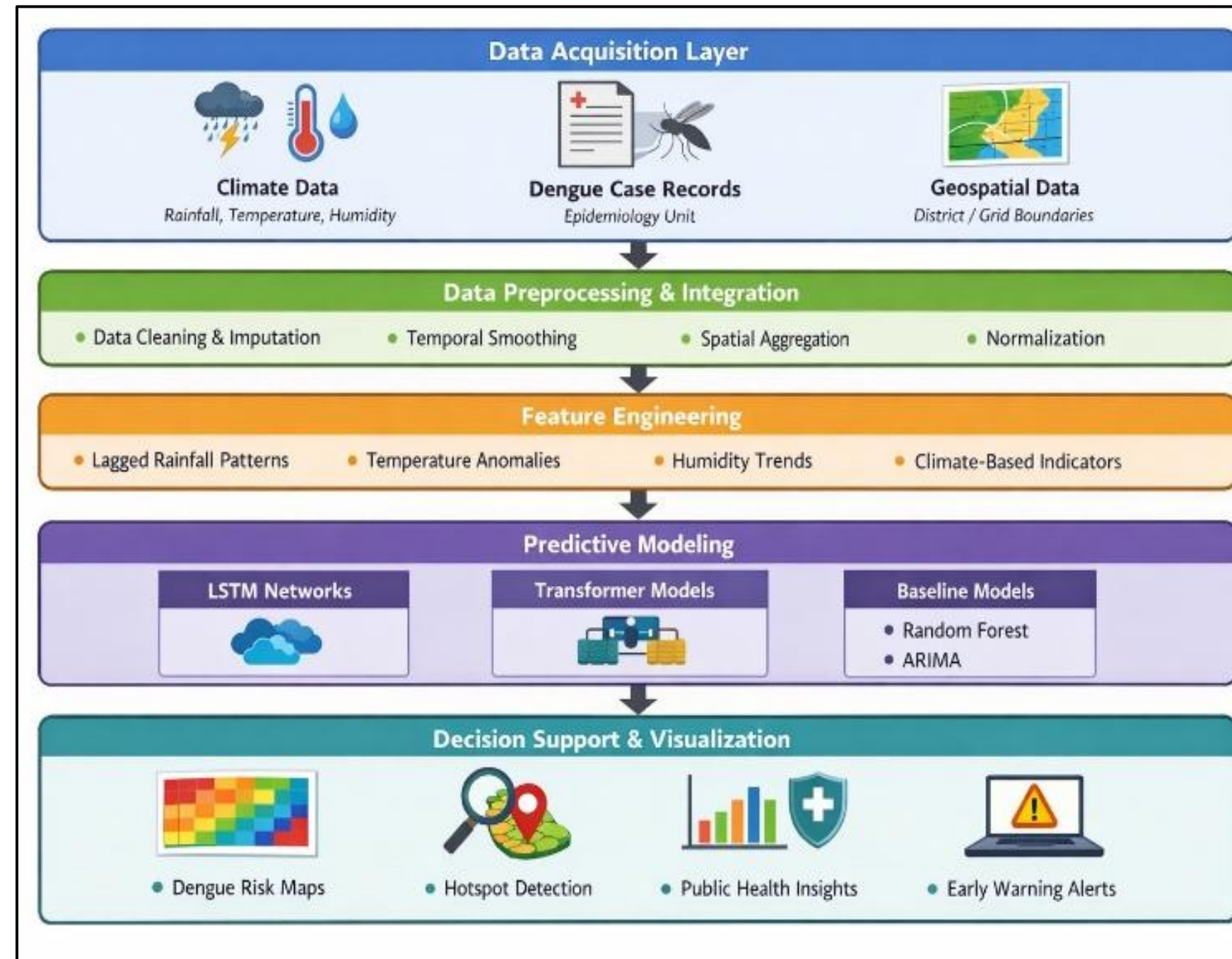
AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

Datasets Used

Dataset	Source	Used In Component	Purpose
Dengue Dataset with Weather Data in Sri Lanka (2019-2021)	Kaggle	Data Preprocessing, Feature Engineering, Prediction Model	Combine dengue cases with rainfall, temperature, and humidity for climate-based dengue prediction
Sri Lanka Dengue Cases Dataset (2010-2020)	Kaggle	Historical Trend Analysis, Model Training, Risk Evaluation	Learn long-term dengue trends, seasonality, and outbreak history in Sri Lanka
Rainfall, Temperature, humidity Data	Department of Meteorology	LSTM Model	Detect spatial Rainfall, Temperature, humidity patterns

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

System Overview



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IT22299666 | ATHUKORALA C.J

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

Problem definition

- Paddy farming is affected by climate changes.
- Farmers mainly depend on traditional knowledge or general weather forecasts.
- Irrigation is sometimes done at the wrong time. Fertilizers are not always applied properly.

Research objectives

- Predict paddy yield using historical climate and agricultural data via machine learning.
- Advise farmers on irrigation and fertilizer timing based on climate insights.
- Deliver recommendations through a simple, accessible digital platform

Novelty

- Existing research that only predicts yield numbers, this system converts predictions into simple language and advises farmers on what actions to take next
- Existing research uses irrigation based on weather forecasts, which helps with water management. However, it does not support fertilizer advice and crop management. This system addresses both.

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

Research Methodology

- Uses Design Science Research (design, implementation, evaluation).
- Uses climate data (rainfall, temperature) and paddy yield records.
- Python pipeline for data cleaning and preprocessing.
- Random Forest model to identify key climate features and yield patterns.
- LSTM model to learn time-based climate trends and predict yield.

Latest Technologies

- Python, Pandas, NumPy, and SciPy were used for data processing.
- Random Forest was used to select important features and classify yield conditions.
- LSTM model was developed for sequential climate and yield prediction.
- GIS was used for district-level climate visualization.
- React Native was used to build the mobile app.
- Firebase was used for database, authentication, and data storage.

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

Validation & Evaluation

- MAE, RMSE – prediction accuracy
- Precision, Recall, F1-score – recommendation quality
- Comparison with Random Forest baseline
- Farmer usability feedback

Data, Security & Ethics

- Data from Sri Lankan government agencies
- DoA fertilizer guidelines used as advisory rules
- Firebase security & HTTPS protection
- Voluntary farmer participation

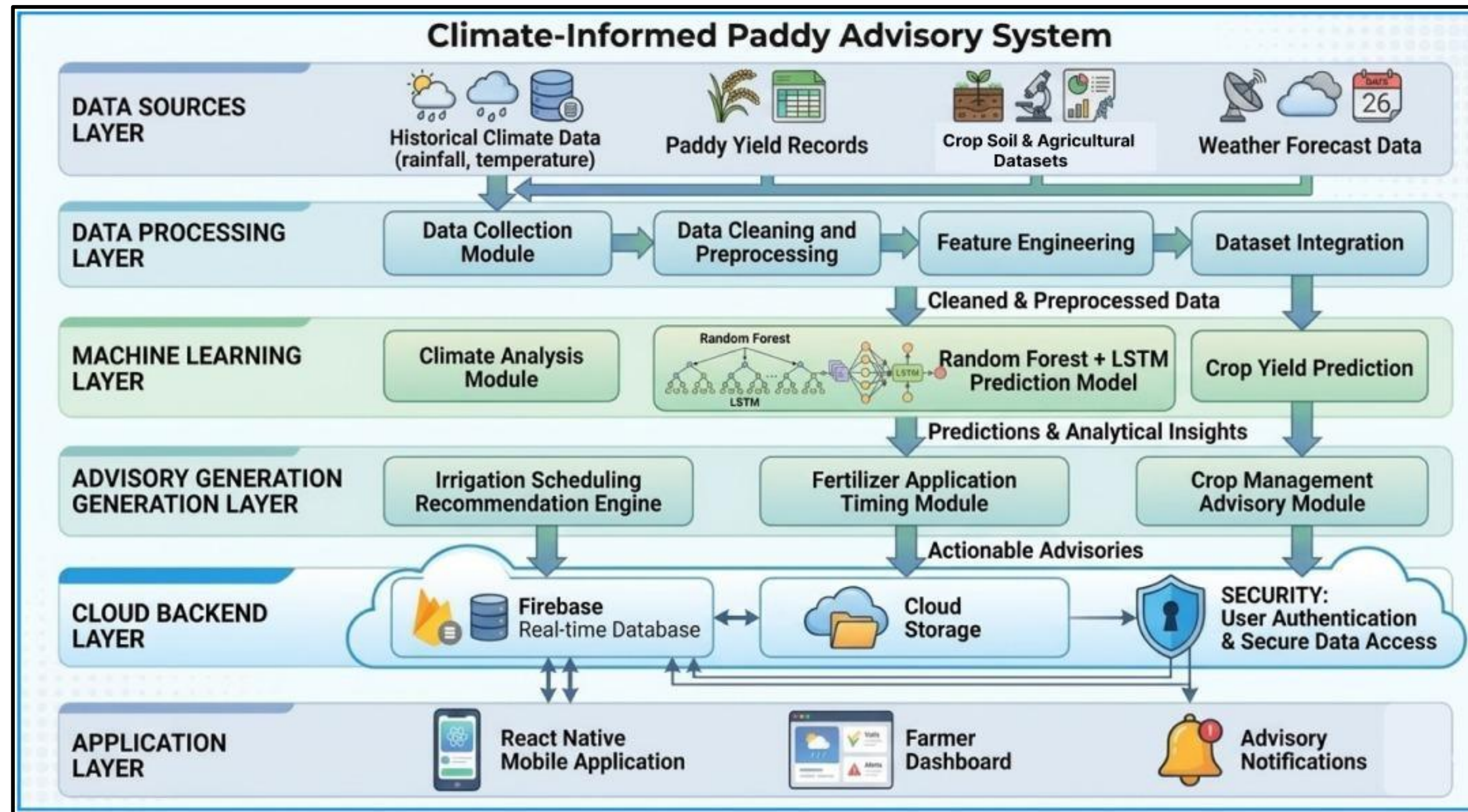
CLIMATE-INFORMED PADDY ADVISORY SYSTEM

Datasets Used

Dataset	Source	Used In Component	Purpose
Historical Paddy Yield Data	Department of Agriculture (Sri Lanka)	Data Preprocessing & Feature Engineering ML Predictive Model (CNN-LSTM)	Train ML models to predict paddy yield patterns across regions and seasons.
Climate Data	Department of Meteorology Sri Lanka	Data Preprocessing & Feature Engineering ML Predictive Model: ML Predictive Model:	Link rainfall, temperature, humidity, and other climatic variables with crop growth and yield outcomes.
Crop & Soil Data	Kaggle	Irrigation scheduling module, fertilizer application timing, yield prediction	Track growth stages

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

System Overview





IT22553232 | ABHISHEKA K.K.R.H

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

Problem definition

- Heavy rain and floods in Sri Lanka often make roads unsafe.
- Current navigation apps (Google Maps) focus on traffic and shortest path only.
- They do not consider weather risks like flooding or heavy rain.
- Drivers may get fast routes but unsafe routes during bad weather.
- Most systems use only traffic data, not multiple data sources.

Research objectives

- Develop an AI-based climate-aware routing system.
- Use traffic, weather, and road incident data.
- Predict road hazards using LSTM machine learning model.
- Generate safer routes.
- Provide real-time safe navigation through a mobile app.

Novelty

- Climate-aware routing (considers weather risks).
- Hazard prediction using AI (LSTM).
- Combines traffic, weather and incident data.
- Risk-based route selection, not only shortest path.
- Designed specifically for Sri Lankan road and climate conditions.

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

Research Methodology

- Uses Design Science Research (design, implementation, evaluation).
- Temporal hazard prediction from multi-source time-series data
- Graph-based routing optimizing travel time and environmental risk scores

Latest Technologies

- Python – Data processing
- TensorFlow – Machine learning model development
- Firebase & NoSQL Database – Real-time data storage
- React Native – Mobile application development

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

Validation & Evaluation

- Measures hazard prediction accuracy using Precision, Recall & F1-score
- Evaluates hazard vs non-hazard classification Route comparison
- Climate-aware routes vs traditional shortest path routes

Data, Security & Ethics

- Weather from OpenWeather API
- Crowd-sourced incident reports, HTTPS encryption and Firebase access control protect all user and route data.
- No personally identifiable information collected

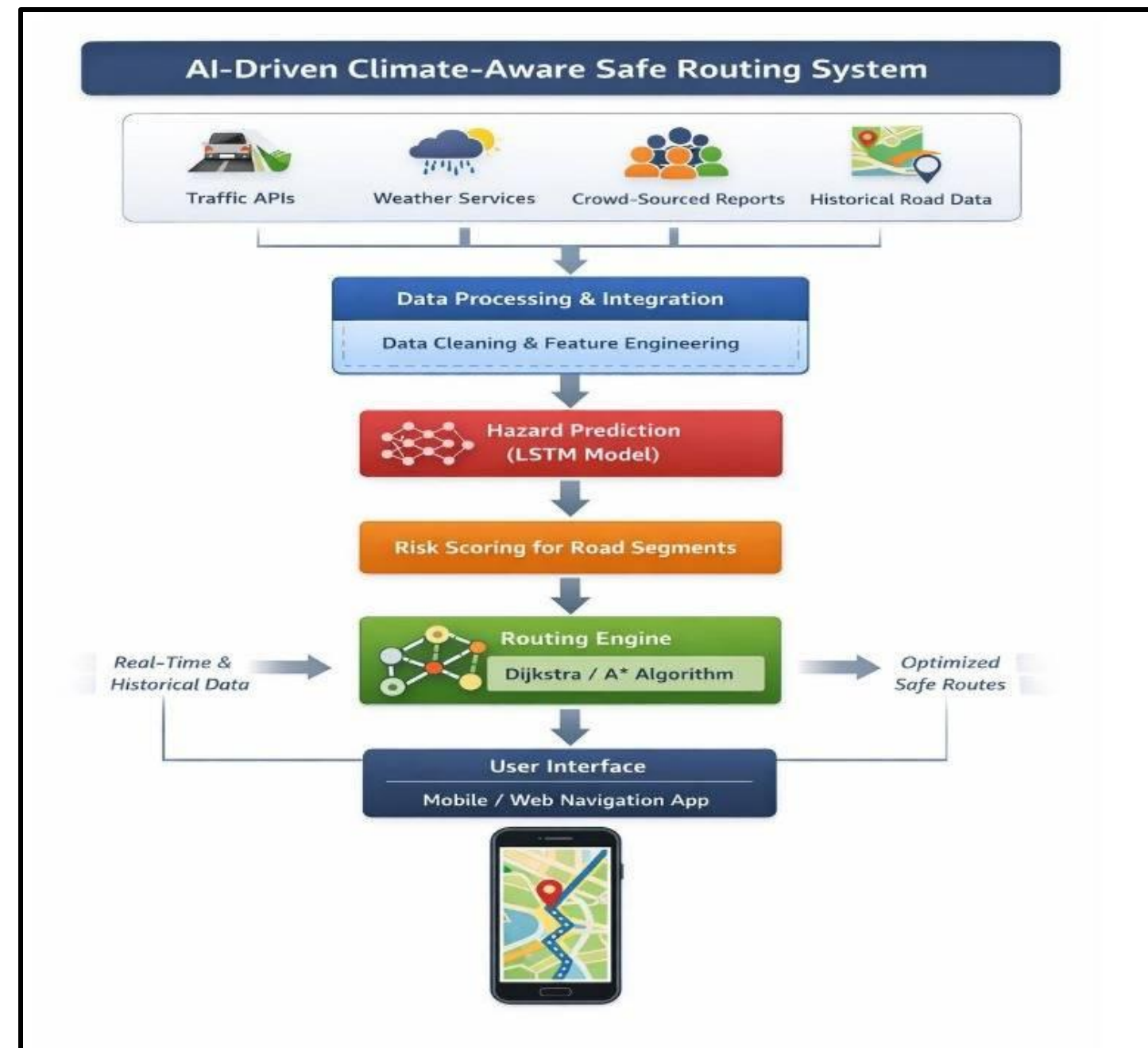
AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

Datasets Used

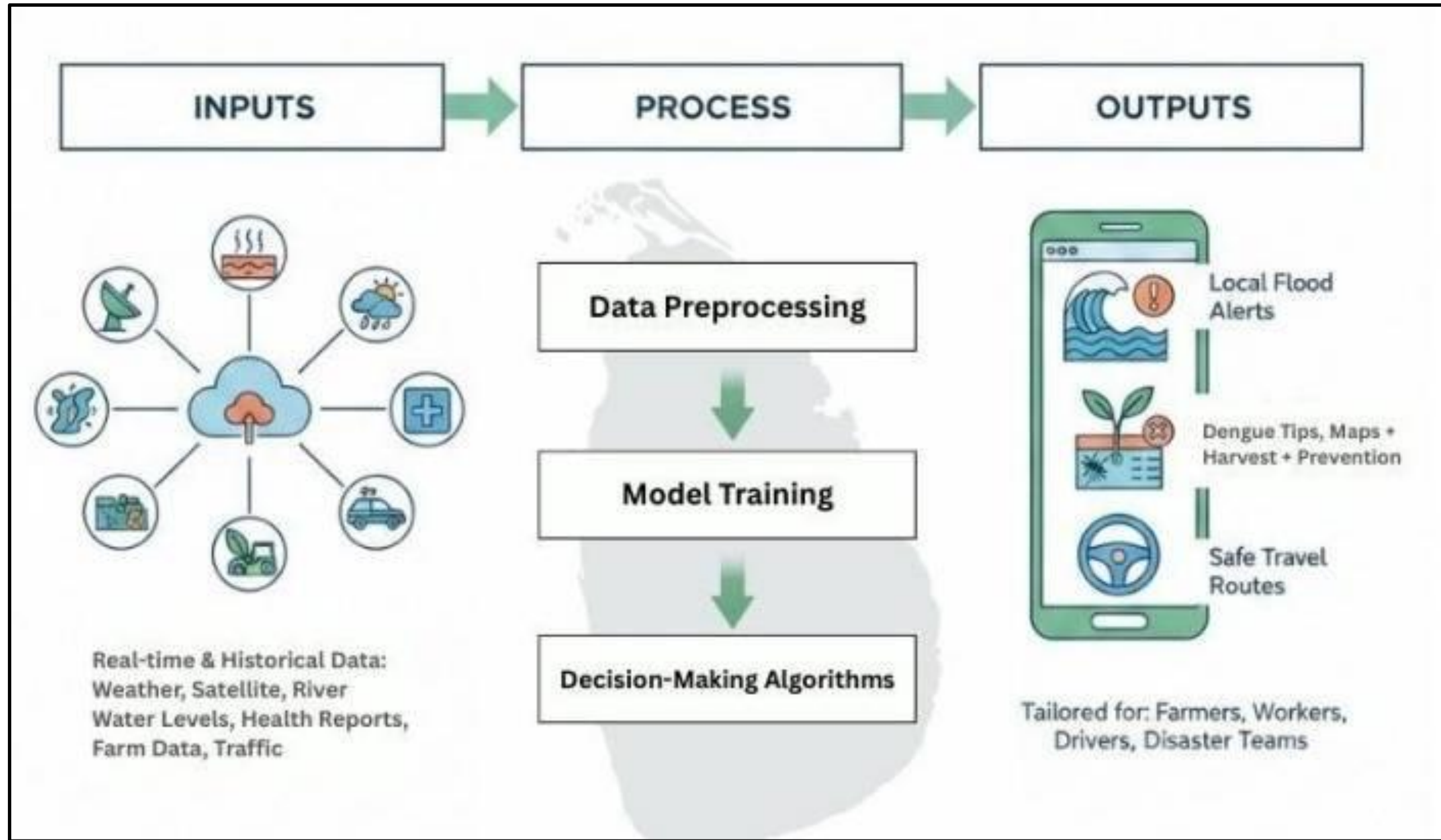
Dataset	Source	Used In Component	Purpose
Weather Data (Temperature, Humidity, Rainfall)	Sri Lanka Weather Dataset (Kaggle – 2010-2023)	ML Model Training	Train predictive models using long-term weather observations across Sri Lankan cities.
Sri Lanka Flood Risk & Inundation Dataset	Kaggle	Data Preprocessing & Feature Engineering; Hazard Prediction Module (LSTM Model)	Identify flood-prone regions and analyse historical flood patterns to help the system predict road hazards and assign risk scores to road segments during heavy rainfall events.

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

System Overview



SYSTEM DIAGRAM



MARKET, VALUE PROPOSITION & COMMERCIALIZATION

Ability of Commercialization

- High national demand for disaster and climate alerts
- Supports government agencies and public safety
- Scalable across Sri Lanka and South Asia

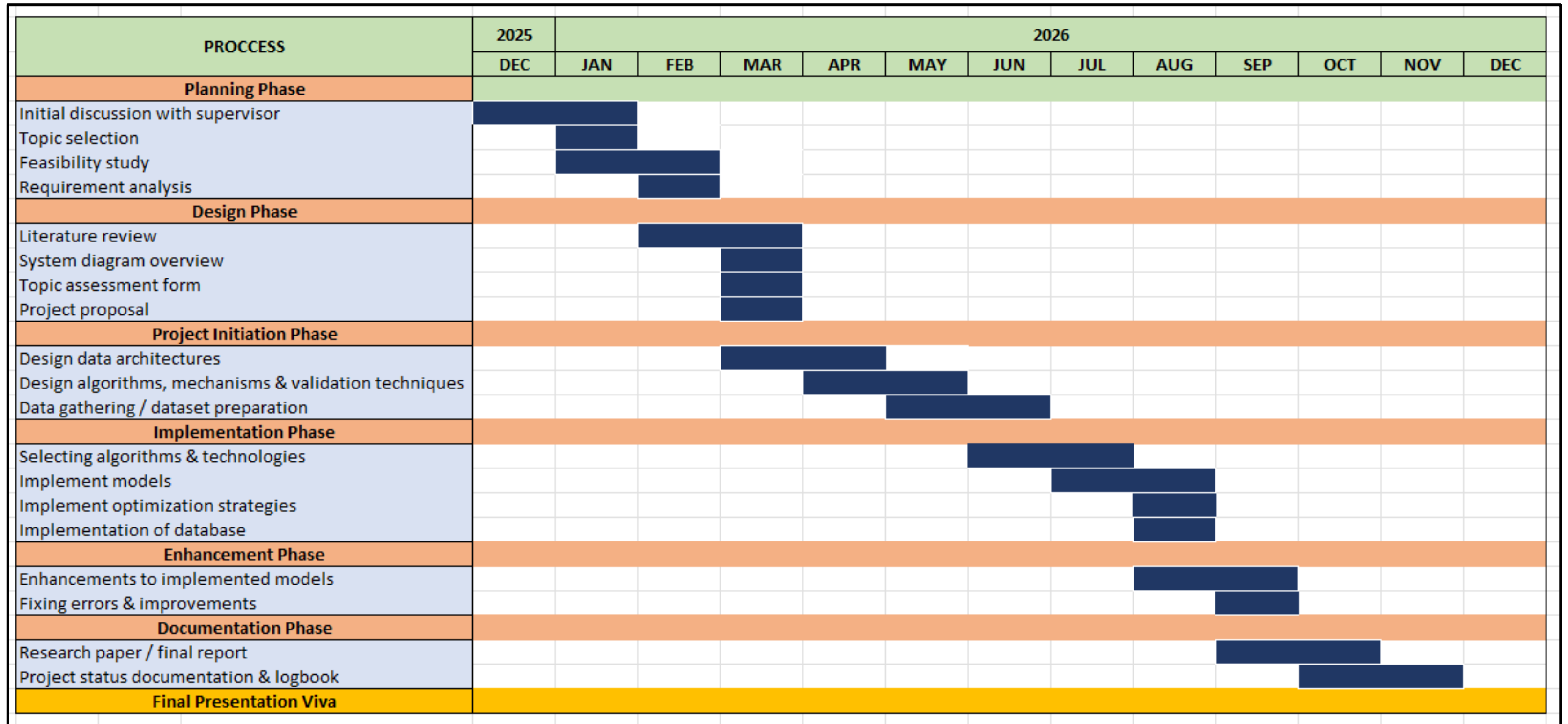
Value Proposition

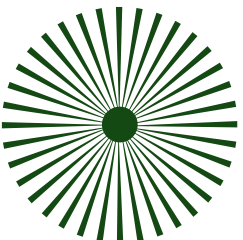
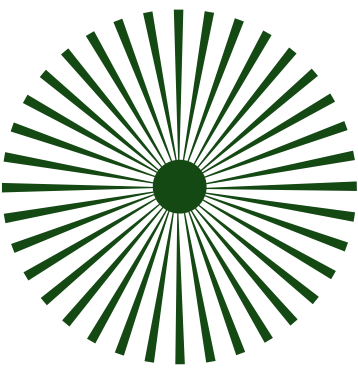
- Farmers: Accurate climate-based farming decisions
- Government: Wider disaster monitoring without new infrastructure
- Health Authorities: Early dengue outbreak detection
- Society: AI-based prevention for national risks

Commercialization Pathway / Revenue Streams

- Primary: Institutional licensing to government agencies
- Secondary: Climate analytics and disaster planning reports
- Tertiary: White-label licensing to municipalities, NGOs, insurers, and agribusinesses

GANTT CHART





THANK YOU ...

